

TaskBench: Measuring Cost-Quality Tradeoffs Across 46 LLMs and 21 Production Tasks

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Abstract

We benchmark 46 large language models from 12 API providers across 21 production tasks spanning sentiment classification, intent detection, code generation, mathematical reasoning, and 17 others. We measure both per-query cost and quality on 50 cases per task (51,580 scored cases total). Economy-tier models (\$0.05–\$0.49/M input tokens) show no statistically significant quality gap versus Premium models (\$5+/M) on 20 of 21 tasks (Mann-Whitney $p > 0.05$; $n_{\text{Premium}} = 2$). Within providers, paying 150× more buys at most +0.13 quality points on a 1–5 scale. Per-task routing to the cheapest adequate model reduces cost by 99.7% versus the best single model with negligible quality loss. Quality scores are produced by an LLM judge validated against ground truth on three task types ($r = 0.89$ – 0.91). We release the full dataset, raw model responses, and evaluation prompts.¹

1 Introduction

1.1 Motivation

Practitioners choose large language models by reputation or leaderboard rank. Existing benchmarks measure capability along a single axis: MMLU [Hendrycks et al., 2021] tests academic knowledge, Chatbot Arena [Chiang et al., 2024] aggregates human preference into a single Elo score, and HELM [Liang et al., 2023] evaluates multiple metrics but not cost. A model ranked first on Arena may cost 100× more than the fifth-ranked model for identical quality on sentiment classification. No public dataset lets a practitioner answer a concrete question: for this specific task, which model gives adequate quality at the lowest cost?

1.2 Contributions

We make three contributions.

1. **TaskBench**, a public benchmark measuring per-task cost-quality Pareto frontiers across 46 models, 21 production tasks, 12 providers, and 4 price tiers (51,580 scored cases).
2. **Empirical findings**. Economy-tier models show no significant quality gap versus Premium on 20 of 21 tasks. Per-task routing saves 99.7% versus the best single model. Within-provider price gradients are nearly flat.

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¹Data and code: <https://github.com/SebConejo/benchmark-reverse-engineering>

3. **Judge validation.** We show that standard LLM-as-judge evaluation with generic rubrics has format bias ($r = 0.39$ versus ground truth), fix it with correctness-focused rubrics ($r = 0.91$), and release both score versions for reproducibility.

We release all raw responses, scores, and evaluation code.

1.3 Summary of results

Four headline results:

- *No Economy–Premium gap detected.* Mann-Whitney $p > 0.05$ on 20 of 21 tasks (the single significant result favors Premium on instruction following). Premium averages 4.791 versus Economy 4.698, a gap of 0.093 on the 1–5 scale.
- *Flat provider gradients.* Anthropic Haiku (\$0.80/M) scores 4.775 while Opus (\$15/M) scores 4.753. Google Flash (\$0.15/M) matches Pro (\$1.25/M). OpenAI’s 150× price range yields +0.13 quality.
- *Per-task routing saves 99.7%.* The cheapest adequate model per task averages \$0.000009/query versus \$0.003332 for o3 alone, with at most 0.2 points quality difference on any task.
- *Five Economy models cover all tasks.* At the 4.0/5 quality threshold, five models cover all 21 tasks. The cheapest is Grok 4 Fast at \$0.000435/query average.

2 Related Work

2.1 General LLM benchmarks

HELM [Liang et al., 2023] evaluates 30 models across 42 scenarios on 7 metric families, but cost is not among them. Chatbot Arena [Chiang et al., 2024] collects over 1.5 million human preference votes, yet produces a single aggregate ranking without per-task granularity. MMLU [Hendrycks et al., 2021] and its successor MMLU-Pro test academic knowledge rather than production tasks. The Berkeley Function Calling Leaderboard covers a single task type. TaskBench adds cost as a first-class evaluation axis and provides per-task granularity across 21 diverse production tasks.

2.2 Cost-aware LLM evaluation and routing

FrugalGPT [Chen et al., 2023] introduced LLM cascading for cost reduction but evaluated on one task with three models. RouterBench and RouteLLM benchmark router systems, not the underlying models. RouterArena [Lu et al., 2025] evaluates 14 routers on 8,400 queries but does not provide per-task cost-quality data for the models themselves. Artificial Analysis tracks speed and aggregate pricing but not per-task quality. We produce the underlying per-task cost-quality data that any routing system can consume.

2.3 LLM-as-Judge

MT-Bench [Zheng et al., 2023] established LLM-as-judge evaluation. Known biases include position bias, verbosity bias [Saito et al., 2023], and self-enhancement [Panickssery et al., 2024]. We identify a specific format bias that distorts cost-quality comparisons: generic rubrics reward concise formatting over correctness, systematically favoring Economy models. We quantify this ($r = 0.39 \rightarrow 0.91$) and release both V1 and V2 scores.

3 Methodology

3.1 Benchmark design

We evaluate 21 production tasks in four categories, with 50 cases each drawn from standard datasets:

- **Classification** (4 tasks, exact-match): sentiment (SST-2), intent detection (CLINC-150, 150 classes), content moderation (ToxiGen), and multistep reasoning (ARC-Challenge).
- **Structured output** (5 tasks, LLM-judge): entity extraction, JSON transformation, named-entity recognition, function calling, and structured output generation.
- **Generation** (8 tasks, LLM-judge): code generation (HumanEval), code review, test generation, summarization (email and long-form), data-to-text, code explanation, and instruction following.
- **Reasoning and language** (4 tasks, LLM-judge): mathematical reasoning (GSM8K), extractive QA (SQuAD v2), translation EN→FR (OPUS-100), and SQL generation (Spider).

Each case has a fixed prompt sent as a single user message with no system prompt. Models receive identical inputs with temperature = 0 (or 1 for models that reject zero temperature). Two tasks have fewer than 50 cases due to dataset size: code explanation (48) and structured output (41).

3.2 Model selection

We evaluate 46 models that completed all 21 tasks with at least 40 valid cases each, drawn from 12 API providers. We assign models to four price tiers based on input token price:

Tier	Price range (input \$/M)	n
Premium	$\geq$ \$5.00	2
Standard	\$0.50–\$4.99	18
Economy	\$0.05–\$0.49	23
Micro	$<$ \$0.05	3

The two Premium models are Claude Opus 4.7 (\$15/M) and GPT-5.5 Pro (\$15/M). Providers include OpenAI (10 models), Qwen (7), ByteDance (5), Mistral (5), Anthropic (4), Google (4), xAI (3), Meta (3), DeepSeek (2), MiniMax (1), Moonshot (1), and Microsoft (1).

Cost is measured as per-query cost from token counts multiplied by provider-published prices at benchmark time (April–May 2026). Reasoning model costs were verified against provider billing dashboards where possible. See Section 6 for the GPT-5.5 Pro cost discrepancy.

3.3 Scoring

We use two evaluation methods:

Exact match (4 classification tasks). Model output is compared to the gold label after lowercasing and whitespace normalization. Score: 1 (correct) or 0 (incorrect), averaged to accuracy per model per task.

LLM-as-judge (17 generative tasks). GPT-4o evaluates each response on a 1–5 integer scale using task-specific correctness rubrics. Each rubric defines what scores of 1, 3, and 5 mean for that specific task. All rubrics include the anti-format instruction: “Ignore formatting, verbosity, and style. Evaluate only correctness and completeness.”

Statistical methods. Bootstrap 95% confidence intervals (1,000 iterations, seed 42). Mann-Whitney U tests for tier comparisons. Kruskal-Wallis tests for multi-group comparisons.

3.4 Judge validation

We validate the LLM judge against ground truth on three task types.

Math reasoning (GSM8K). We compare judge scores to numerical accuracy (does the model’s final answer match the expected number?). With generic V1 rubrics (GPT-4o-mini judge), $r = 0.388$. With correctness-focused V2 rubrics (GPT-4o judge), $r = 0.905$. The V1 judge evaluated “reasoning quality” and penalized verbose format. Claude Opus (96% accuracy) scored 3.48/5 under V1 because of its verbose chain-of-thought. V2 evaluates whether the final answer is correct regardless of presentation.

Extractive QA (SQuAD v2). Judge scores versus exact-match accuracy. V1: $r = -0.013$ (zero correlation). V2: $r = 0.888$. The V1 judge rewarded fluent hallucinations over terse correct answers.

Code generation (HumanEval). Judge scores versus pass@1 execution rate. V2: $r = 0.891$. A naive test harness yielded only $r = 0.49$ due to indentation extraction failures; a multi-strategy harness resolved this (see Appendix C).

The V1-to-V2 improvement is driven by rubric specificity, not the judge model upgrade. V1 evaluated “quality” and rewarded concise formatting. V2 evaluates correctness and penalizes wrong answers regardless of style. The bias changed 34.7% of pairwise model rankings across all 17 LLM-judged tasks (5,195 of 14,972 pairs flipped).

We applied V2 rubrics to all 17 LLM-judged tasks (40,350 cases re-scored at approximately \$60). The 14 tasks without ground truth are validated by mechanism consistency (identical rubric structure and judge model), not by direct ground-truth correlation. This is a limitation we discuss in Section 6.

4 Results

4.1 Cost-quality landscape

Figure 1 shows the 46×21 benchmark matrix. The dominant pattern is uniformity: most cells score above 4.5/5. Quality variance comes from 10 high-discrimination tasks where score spread exceeds 1.5 across models. On these tasks, model choice matters.

The five most discriminative tasks are multistep reasoning (spread 3.68), intent detection (CLINC-150, spread 3.20, $\sigma = 0.624$), extractive QA (spread 2.84, $\sigma = 0.560$), content moderation (spread 2.40), and test generation (spread 2.40). The least discriminative is structured output (spread 0.29).

For 11 of 21 tasks, the score spread is below 1.5 and model selection has minimal impact on quality. On these tasks, the practical recommendation is to use the cheapest available model.

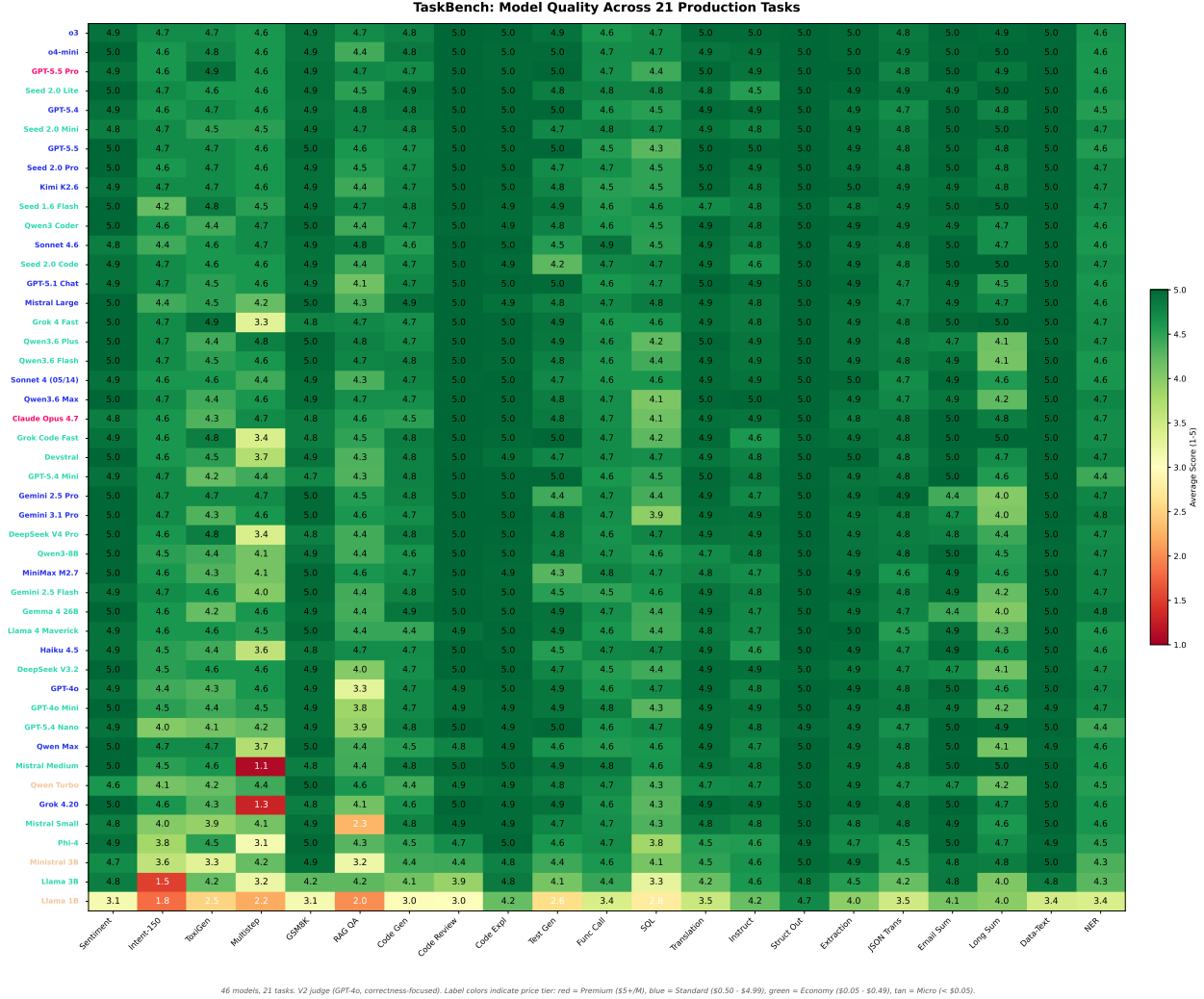


Figure 1: Quality heatmap (46 models $\times$ 21 tasks). Rows sorted by average score, columns grouped by task category. Color intensity indicates mean score per (model, task) pair on the 1–5 scale. Most cells are dark (above 4.5), with visible variation concentrated in classification and QA tasks.

4.2 Tier comparison

Bootstrap 95% confidence intervals per tier (aggregated across all tasks):

Tier	Mean	95% CI	n (pairs)
Premium	4.791	[4.729, 4.853]	42
Standard	4.756	[4.722, 4.785]	378
Economy	4.698	[4.665, 4.731]	483
Micro	4.104	[3.904, 4.303]	63

The Premium-to-Economy gap is 0.093 points (2.3% of the 1–5 scale). Confidence intervals overlap heavily. Mann-Whitney tests find no significant difference on 20 of 21 individual tasks ($p > 0.05$). The single significant result is instruction following, where Premium outperforms Economy ($p = 0.050$). With 21 simultaneous tests at $\alpha = 0.05$, one rejection is expected by chance.



Figure 2: Average quality by price tier with bootstrap 95% confidence intervals. Premium and Economy intervals overlap on all 21 tasks. Micro is the only tier with a visible quality gap.

Micro is the only tier with a large gap (-0.687 versus Premium). This is driven by the three Micro models, which include the two sub-3B models in our sample (Llama 1B and Ministral 3B).

Caveat. With $n = 2$ Premium models, the Mann-Whitney test has low statistical power. We report “no difference detected,” not “tiers are equal.” A future benchmark with five or more Premium models could detect a small gap. Even if a gap exists, it is bounded below 0.14 (the difference between the upper CI of Premium and the lower CI of Economy).

4.3 Per-task Pareto frontiers and routing savings

For each task we compute the cost-quality Pareto frontier: the set of models where no other model is both cheaper and higher quality. Premium models are almost never on the frontier. The frontier is dominated by Economy and Micro models.

Per-task routing to the cheapest model that matches o3 quality on each task saves 99.7% (\$0.003332/query for o3 alone versus \$0.000009/query for the routed portfolio). Even against the cheapest single model covering all 21 tasks at the 4.5 threshold (Grok 4 Fast at \$0.000435/query), routing saves approximately 98%.

A portfolio of just four models covers all 21 tasks at the 4.0 quality threshold: GPT-5.4 Nano on 16 tasks, Phi-4 on 3, Qwen Turbo on 1, and Gemma 26B on 1.

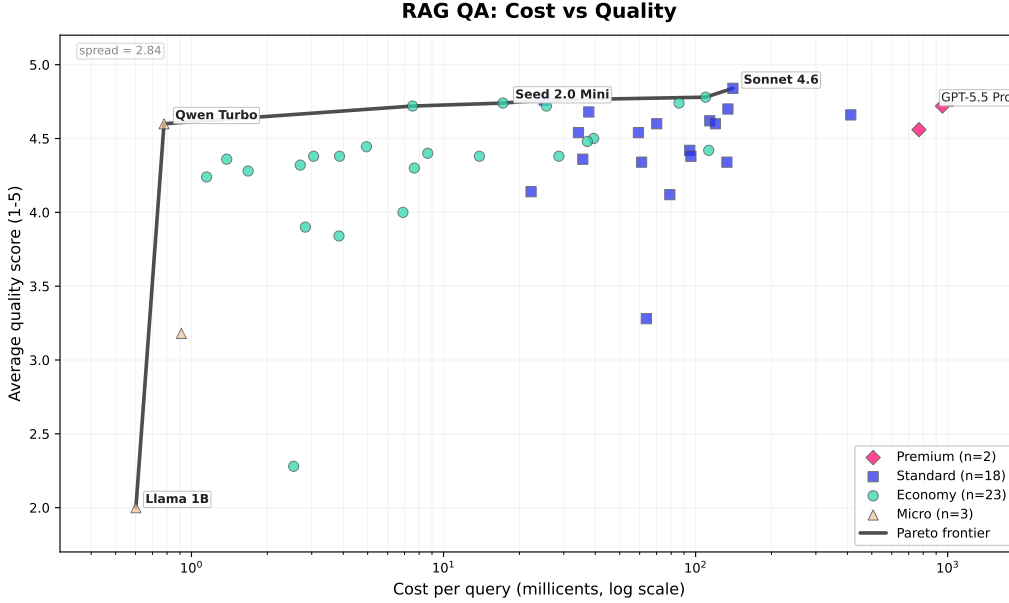
4.4 Within-provider price gradient

Within each major provider, the most expensive model is not reliably the best.

Anthropic. Haiku at \$0.80/M scores 4.775. Opus at \$15/M scores 4.753. Paying $19\times$ more yields -0.022 quality.

Google. Flash at \$0.15/M scores 4.749. Pro at \$1.25/M scores 4.747. An $8\times$ price increase yields -0.002 .

OpenAI. Nano at \$0.10/M scores 4.694. GPT-5.5 Pro at \$15/M scores 4.828. A $150\times$ price increase yields $+0.134$, the largest intra-provider gain in our data, and still only 2.7% of the scale.



n = 46 models, 45-50 cases per model. X-axis: log scale. Black line: Pareto frontier. Tier colors: red = Premium (\$5+/M), blue = Standard (\$0.50 - \$4.99), green = Economy (\$0.05 - \$0.49), tan = Micro (< \$0.05).

Figure 3: Pareto frontier for extractive QA (high-discrimination task, log-scale cost axis). Sonnet 4.6 dominates at 0.14mc; GPT-5.5 Pro costs 7 $\times$ more for lower quality. Black line marks the Pareto frontier. Point shapes and colors indicate price tiers (red = Premium, blue = Standard, green = Economy, tan = Micro).

Qwen. Turbo at \$0.05/M scores 4.654. Max at \$2.00/M scores 4.746. A 40 $\times$ price increase yields +0.092.

Mistral. Ministral 3B at \$0.04/M scores 4.441. Large at \$2.00/M scores 4.800. A 50 $\times$ price increase yields +0.359, the only provider where paying more substantially helps. This is driven by the sub-3B model floor: Ministral 3B is the smallest model in our benchmark.

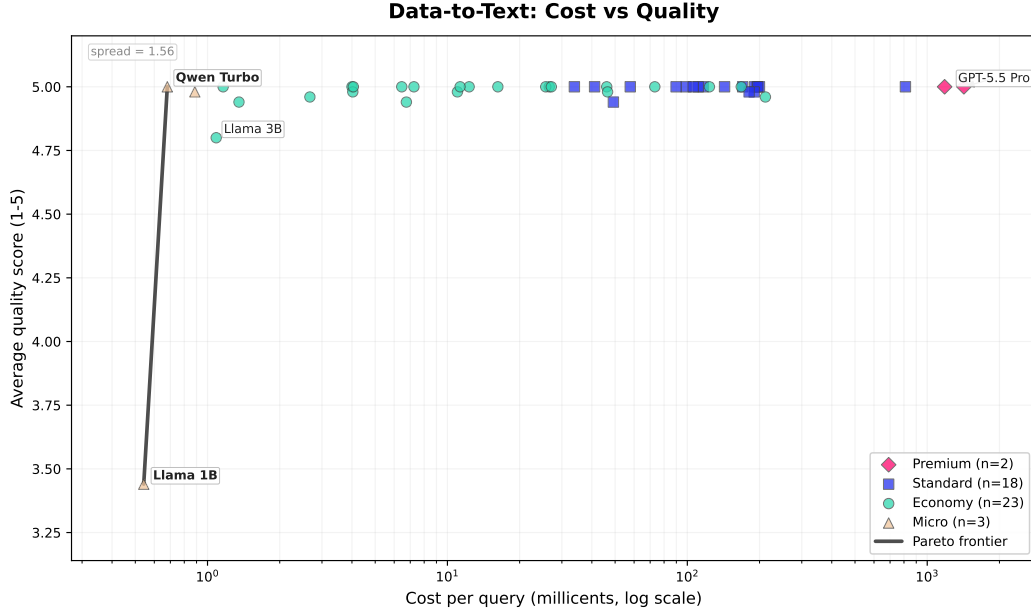
The gradient is flat at the Standard/Economy boundary. The only steep segment is from Micro to Economy, driven by sub-3B models pulling the Micro average down.

4.5 Model selection in practice

We now consider practical model selection strategies given the flat price-quality gradient.

Coverage thresholds. At the 4.0/5 quality threshold, 37 of 46 models cover all 21 tasks. Model choice barely matters. At the 4.5 threshold, only five models qualify as universal: Grok 4 Fast, Seed 2.0 Mini, GPT-5.4, Seed 2.0 Lite, and o3. The cheapest is Grok 4 Fast at \$0.000435/query average. At the 4.7 threshold, no single model covers all 21 tasks. The bottleneck tasks are extractive QA, intent detection, and content moderation.

Generational upgrades. The cost-quality frontier moves with each model generation. GPT-4o to GPT-5.4: -60% price (\$2.50 $\rightarrow$ \$1.00/M), +0.089 average quality. GPT-4o Mini to GPT-5.4 Mini: -33% price, quality flat (already near ceiling). Sonnet 4 to Sonnet 4.6: same price, +0.003 quality. The exception is DeepSeek V3.2 to V4 Pro: +211% price for +0.061 quality. In three of four



n = 46 models, 50 cases per model. X-axis: log scale. Black line: Pareto frontier. Tier colors: red = Premium (\$5+/M), blue = Standard (\$0.50 - \$4.99), green = Economy (\$0.05 - \$0.49), tan = Micro (< \$0.05).

Figure 4: Pareto frontier for data-to-text (commodity task, log-scale cost axis). Qwen Turbo achieves a perfect 5.0 at 0.0007 mc; 30+ models match that score at up to 2,000 $\times$ the cost. The near-flat frontier illustrates a task where model selection is cost-irrelevant.

provider families, the new generation is cheaper at equal or better quality. Upgrading generations is a more reliable strategy than upgrading tiers.

5 Discussion

5.1 Practical implications

Three recommendations for practitioners:

1. **Default to Economy tier.** On 20 of 21 tasks, Economy models showed no significant quality difference versus Premium (the single exception favors Premium on instruction following). The safest default is the cheapest model scoring at least 4.5 on the target task.
2. **Route per task for maximum savings.** A four-model portfolio covers all 21 tasks at the 4.0 threshold and costs \$0.000009/query average. The savings come from commodity tasks where Micro-tier models suffice.
3. **Upgrade generations, not tiers.** GPT-5.4 is 60% cheaper than GPT-4o at higher quality. The cost-quality frontier moves down-left with each generation. Paying for a premium tier within the same generation has near-zero return.

Premium models may add marginal value on open-ended language tasks, but the effect size (< 0.2 points) is within our task-level confidence intervals.

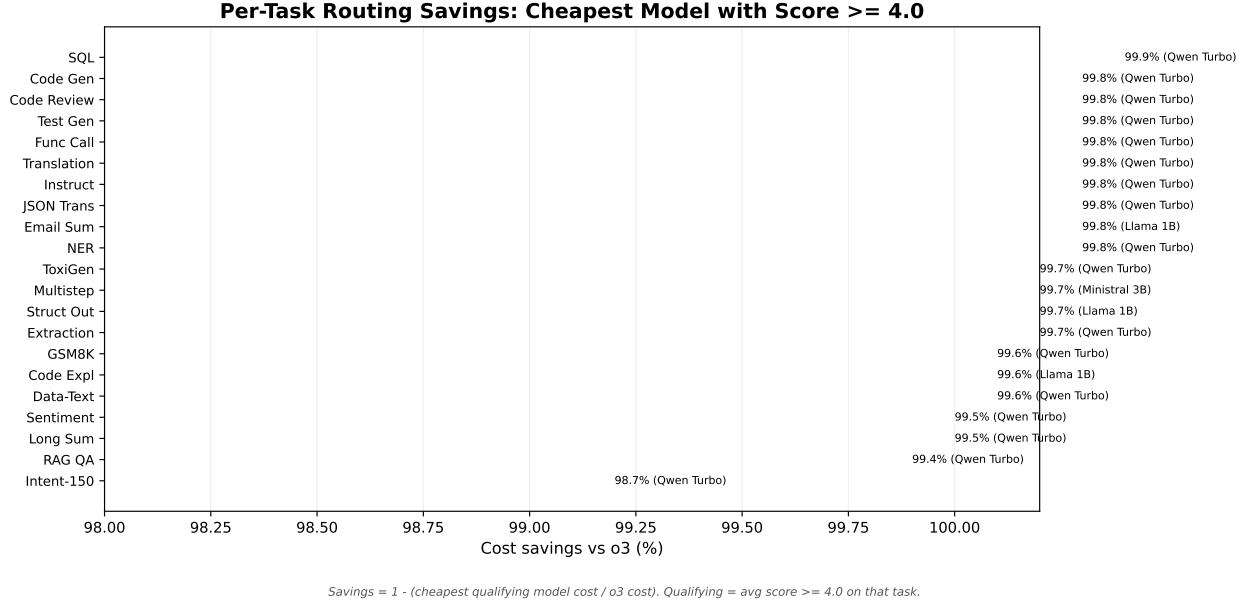


Figure 5: Per-task routing savings versus o3. Each bar shows the cost reduction from routing to the cheapest model scoring ≥ 4.0 on that task. All 21 tasks achieve $> 98.7\%$ savings. The x-axis is zoomed to 98–100% to make inter-task differences visible.

5.2 Evaluation methodology implications

The format bias finding (Section 3.4) generalizes beyond this benchmark. Any LLM-as-judge evaluation using generic “rate quality 1–5” rubrics likely contains this bias. We recommend three practices: (1) always validate against ground truth on at least two task types, (2) use task-specific correctness anchors rather than generic quality rubrics, and (3) report both judge scores and native metrics where available.

Verbosity is not the primary driver of the bias ($r = 0.22$, $p = 0.13$ between verbosity ratio and V1-to-V2 score delta). The mechanism is rubric ambiguity: generic rubrics allow the judge to reward format features that correlate with model price tier.

6 Limitations

- **$n = 2$ Premium models.** The Economy-matches-Premium finding is “no difference detected,” not “tiers are equal.” Two Premium models (Claude Opus 4.7 and GPT-5.5 Pro) provide low statistical power. Future work should include five or more Premium models.
- **Judge validated on 3 of 17 LLM-judged tasks.** The remaining 14 tasks are validated by mechanism consistency (identical rubric structure and GPT-4o judge), not by direct ground-truth correlation.
- **50 cases per task.** Confidence intervals are approximately ± 0.2 . Differences below 0.3 are not distinguishable from noise. This is standard for benchmark papers; HELM uses similar per-scenario sample sizes.
- **Single-turn only.** Multi-turn conversations, vision, and audio inputs are out of scope.

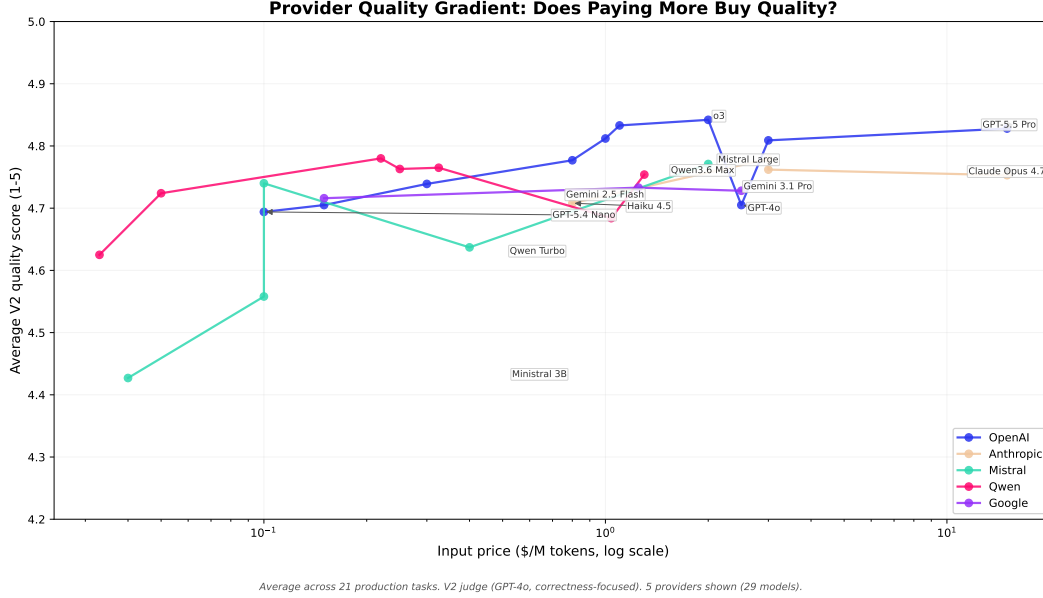
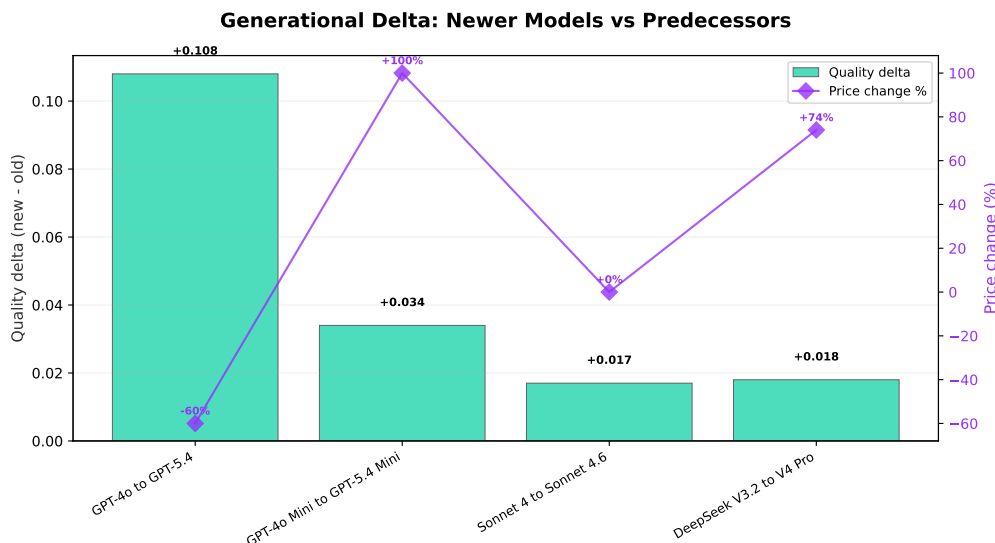


Figure 6: Within-provider price-quality gradient. Each line connects models from the same provider, ordered by input price. The gradient is nearly flat for Anthropic, Google, and Qwen. OpenAI shows the largest absolute gain (+0.13) across its 150 $\times$ price range. Mistral is the exception, driven by its sub-3B entry-level model.

- **Price snapshot (April–May 2026).** LLM prices change monthly. Tier-level findings are more durable than absolute prices because relative pricing between tiers rarely inverts. We release raw data for re-computation with current prices.
- **Potential data contamination.** Models may have seen GSM8K, SQuAD, and HumanEval during training. This affects absolute scores but not relative comparisons, since all models face the same contamination risk on standard datasets.
- **Reasoning model cost tracking.** GPT-5.5 Pro and o3 reasoning tokens are partially invisible in the API response. Tracked cost (\$143.81 total spend) underestimates real spend (approximately \$180+) for these models specifically. Non-reasoning model costs remain accurate.
- **Model diversity.** Our sample includes three geographic origins and both open-weight and closed-source licenses. Neither dimension significantly predicts quality: provider origin shows no effect (Kruskal-Wallis $H = 3.80$, $p = 0.15$), and the open-versus-closed gap (0.191, $p = 0.0016$) is driven by sub-3B open-weight models (see Appendix G).
- **English-only** (except the EN $\rightarrow$ FR translation task).

7 Conclusion

TaskBench provides a systematic per-task cost-quality benchmark for production LLM use. Across 46 models and 21 tasks, we find that the cost-quality curve is flat: on 20 of 21 tasks we detect no significant difference between Economy and Premium tiers, and the single significant result favors Premium ($n_{\text{Premium}} = 2$; more Premium models are needed to confirm this pattern). The practical implication is that practitioners should default to Economy models and use per-task routing for



Quality delta = average score difference across 21 tasks (new model - predecessor). Price change = input price change (%). Pairs selected where both predecessor and successor have complete 21/21 task coverage.

Figure 7: Quality delta and price change for four generational model pairs. In three of four cases, the newer model is cheaper or equal in price at higher quality. Pairs selected where both predecessor and successor have complete 21/21 task coverage.

maximum savings.

We also show that standard LLM-as-judge evaluation has a format bias that specifically distorts cost-quality comparisons. Correctness-focused rubrics fix this with minimal effort.

We release all 51,580 scored cases, 51,705 raw model responses, evaluation prompts, and analysis code. We encourage the community to extend TaskBench with more Premium models, multi-turn tasks, and longitudinal price tracking.

Acknowledgments

Disclosure. This work was conducted by Sébastien Conejo, co-founder of Manifest (<https://github.com/mnfst/manifest>), an open-source LLM model router. Manifest was not used in this benchmark. The routing savings analysis (Section 4.3) applies to any routing system and is computed from the benchmark data using a simple per-task argmin. All code, data, and evaluation prompts are publicly available.²

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A Full Pareto Frontiers (19 Remaining Tasks)

One single-panel log-scale Pareto figure per task, following the style of Figures 3 and 4. RAG QA and Data-to-Text appear in the main text.

B Complete Model Table

C Judge Validation Details

V1 versus V2 correlation summaries for all four validated task types:

#	Task	V1 r	V2 r	Native metric
1	Math reasoning (GSM8K)	0.388	0.905	Numerical accuracy
2	Extractive QA (SQuAD)	-0.013	0.888	Substring match
3	Code generation (HumanEval)	0.491*	0.891	pass@1
4	Translation (OPUS-100)	n/a	0.290	BLEU (single ref.)

*V1 code correlation uses a naive harness; V2 uses the multi-strategy harness.

The code generation validation required a multi-strategy test harness. A naive harness using `textwrap.indent` failed on 47.2% of responses due to indentation mismatches. Our multi-strategy approach tries four extraction strategies in sequence (standalone function extraction, uniform indent, selective indent, raw append), resolving 91.4% of formatting failures and raising the pass rate from 44.9% to 91.4%.

Translation BLEU is inconclusive ($r = 0.29$, weakly significant) due to the metric’s single-reference limitation, not the judge’s quality.

Per-task ranking flip rates range from 20.2% (code review, Kendall $\tau = 0.596$) to 50.3% (extractive QA, $\tau = -0.006$). The aggregate flip rate is 34.7% (5,195 of 14,972 pairs).

Table 1: Complete model leaderboard: all 46 models evaluated on all 21 tasks (≥ 40 cases each). Sorted by average quality score (1–5 Likert).

Rank	Model	Provider	Tier	Input \$/M	Avg Score
1	o3	OpenAI	Standard	2.00	4.842
2	o4-mini	OpenAI	Standard	1.10	4.833
3	GPT-5.5 Pro	OpenAI	Premium	15.00	4.828
4	Seed 2.0 Lite	ByteDance	Economy	0.250	4.816
5	GPT-5.4	OpenAI	Standard	1.00	4.812
6	Seed 2.0 Mini	ByteDance	Economy	0.100	4.810
7	GPT-5.5	OpenAI	Standard	3.00	4.809
8	Seed 2.0 Pro	ByteDance	Standard	0.600	4.806
9	Kimi K2.6	Moonshot	Standard	0.600	4.796
10	Seed 1.6 Flash	ByteDance	Economy	0.075	4.792
11	Qwen3 Coder	Qwen	Economy	0.220	4.780
12	Sonnet 4.6	Anthropic	Standard	3.00	4.779
13	Seed 2.0 Code	ByteDance	Economy	0.300	4.778
14	GPT-5.1 Chat	OpenAI	Standard	0.800	4.777
15	Mistral Large	Mistral	Standard	2.00	4.771
16	Grok 4 Fast	xAI	Economy	0.200	4.769
17	Qwen3.6 Plus	Qwen	Economy	0.325	4.765
18	Qwen3.6 Flash	Qwen	Economy	0.250	4.763
19	Sonnet 4 (05/14)	Anthropic	Standard	3.00	4.762
20	Qwen3.6 Max	Qwen	Standard	1.30	4.754
21	Claude Opus 4.7	Anthropic	Premium	15.00	4.753
22	Grok Code Fast	xAI	Economy	0.200	4.741
23	Devstral	Mistral	Economy	0.100	4.740
24	GPT-5.4 Mini	OpenAI	Economy	0.300	4.739
25	Gemini 2.5 Pro	Google	Standard	1.25	4.733
26	Gemini 3.1 Pro	Google	Standard	2.50	4.728
27	DeepSeek V4 Pro	DeepSeek	Economy	0.435	4.725
28	Qwen3-8B	Qwen	Economy	0.050	4.724
29	MiniMax M2.7	MiniMax	Standard	1.10	4.718
30	Gemini 2.5 Flash	Google	Economy	0.150	4.716
31	Gemma 4 26B	Google	Economy	0.060	4.714
32	Llama 4 Maverick	Meta	Economy	0.150	4.714
33	Haiku 4.5	Anthropic	Standard	0.800	4.708
34	DeepSeek V3.2	DeepSeek	Economy	0.250	4.707
35	GPT-4o	OpenAI	Standard	2.50	4.705
36	GPT-4o Mini	OpenAI	Economy	0.150	4.705
37	GPT-5.4 Nano	OpenAI	Economy	0.100	4.694
38	Qwen Max	Qwen	Standard	1.04	4.684
39	Mistral Medium	Mistral	Economy	0.400	4.637
40	Qwen Turbo	Qwen	Micro	0.033	4.625
41	Grok 4.20	xAI	Standard	1.25	4.593
42	Mistral Small	Mistral	Economy	0.100	4.558
43	Phi-4	Microsoft	Economy	0.065	4.531
44	Ministral 3B	Mistral	Micro	0.040	4.427
45	Llama 3B	Meta	Economy	0.051	4.145
46	Llama 1B	Meta	Micro	0.027	3.261

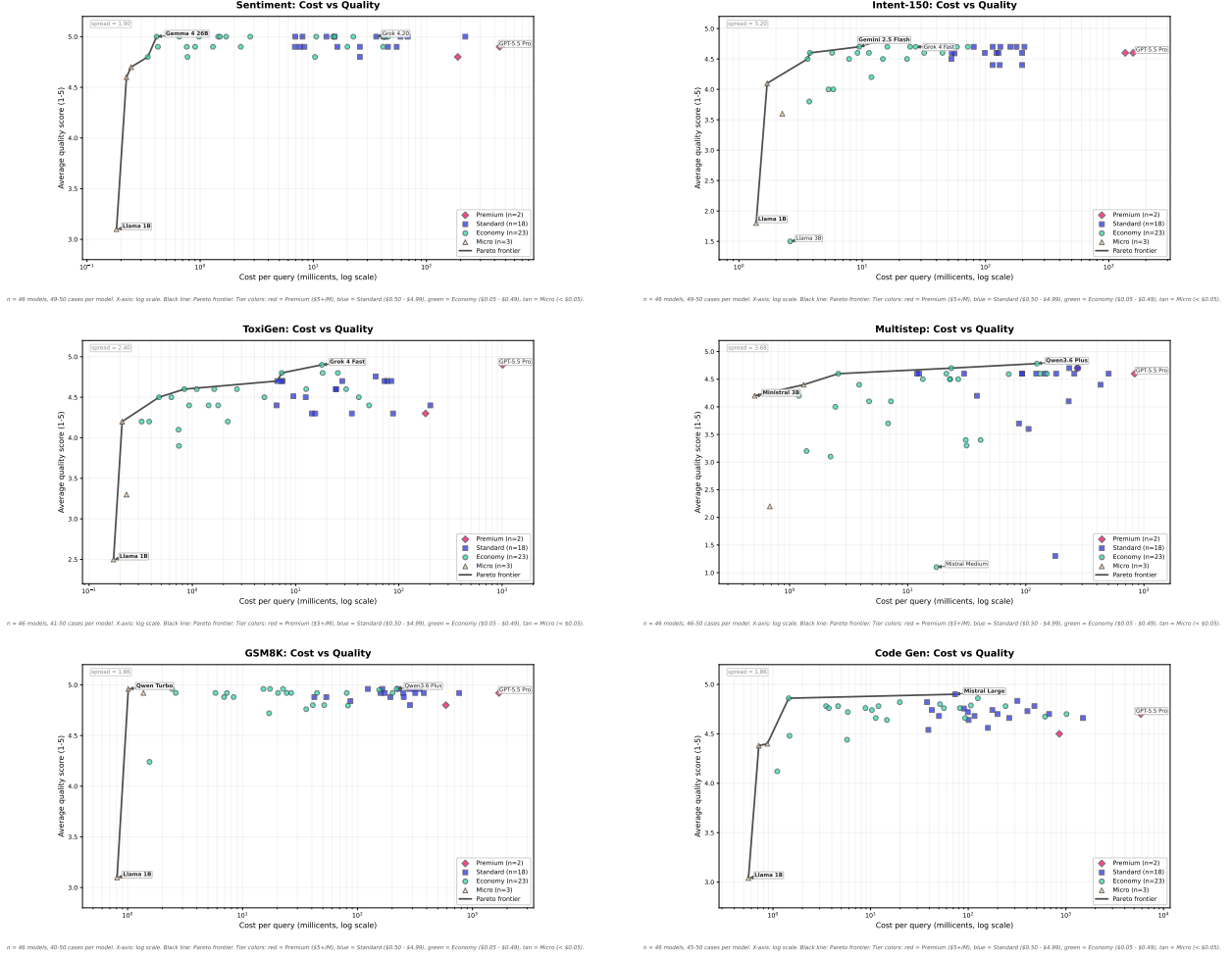


Figure 8: Pareto frontiers (1/4): Classification and reasoning tasks.

D Judge Prompts

Full V1 and V2 judge prompt text for all 17 LLM-judged tasks. Available in `JUDGE_PROMPTS_V2.md` in the companion repository.

E Statistical Tables

Bootstrap 95% confidence intervals per (task, tier) and Mann-Whitney U p -values per task (Economy versus Premium, Economy versus Standard). Coverage matrix showing per-model threshold coverage at 4.0, 4.5, and 4.7. Per-task routing savings (cheapest adequate model, cost, and savings percentage). All data available in `results/analysis_v3_final/` in the companion repository.

F Top-5 Models Per Task

For each of the 21 tasks, the five highest-scoring models ranked by quality (ties broken by lowest cost). Available as `top5_per_task.json` in the companion repository.

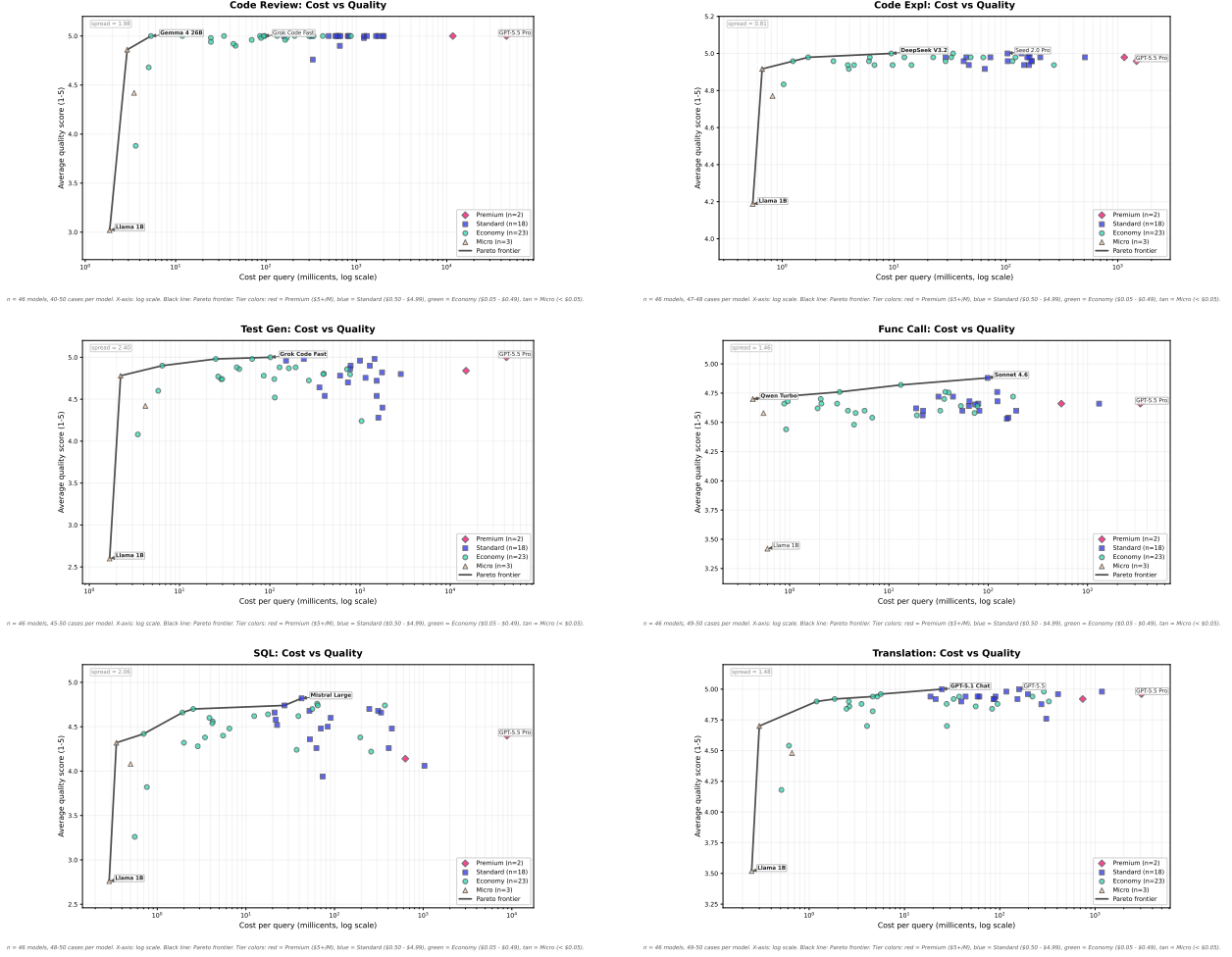


Figure 9: Pareto frontiers (2/4): Code and structured tasks.

G Model Diversity Analysis

Provider origin. Three geographic groups: Chinese ($n = 16$, mean 4.753), American ($n = 25$, mean 4.656), and European ($n = 5$, mean 4.627). Kruskal-Wallis $H = 3.80$, $p = 0.15$. No significant difference. The American group has the widest range (3.261 to 4.842), driven by sub-3B Llama models at the low end.

License type. Open-weight ($n = 17$, mean 4.566) versus closed-source ($n = 29$, mean 4.757). Mann-Whitney $p = 0.0016$, gap 0.191. However, the open-weight group includes all sub-3B models (Llama 1B, Llama 3B, Ministral 3B). Excluding sub-3B models narrows the gap to approximately 0.058. The license effect is driven primarily by model size, not by the open-weight development process.

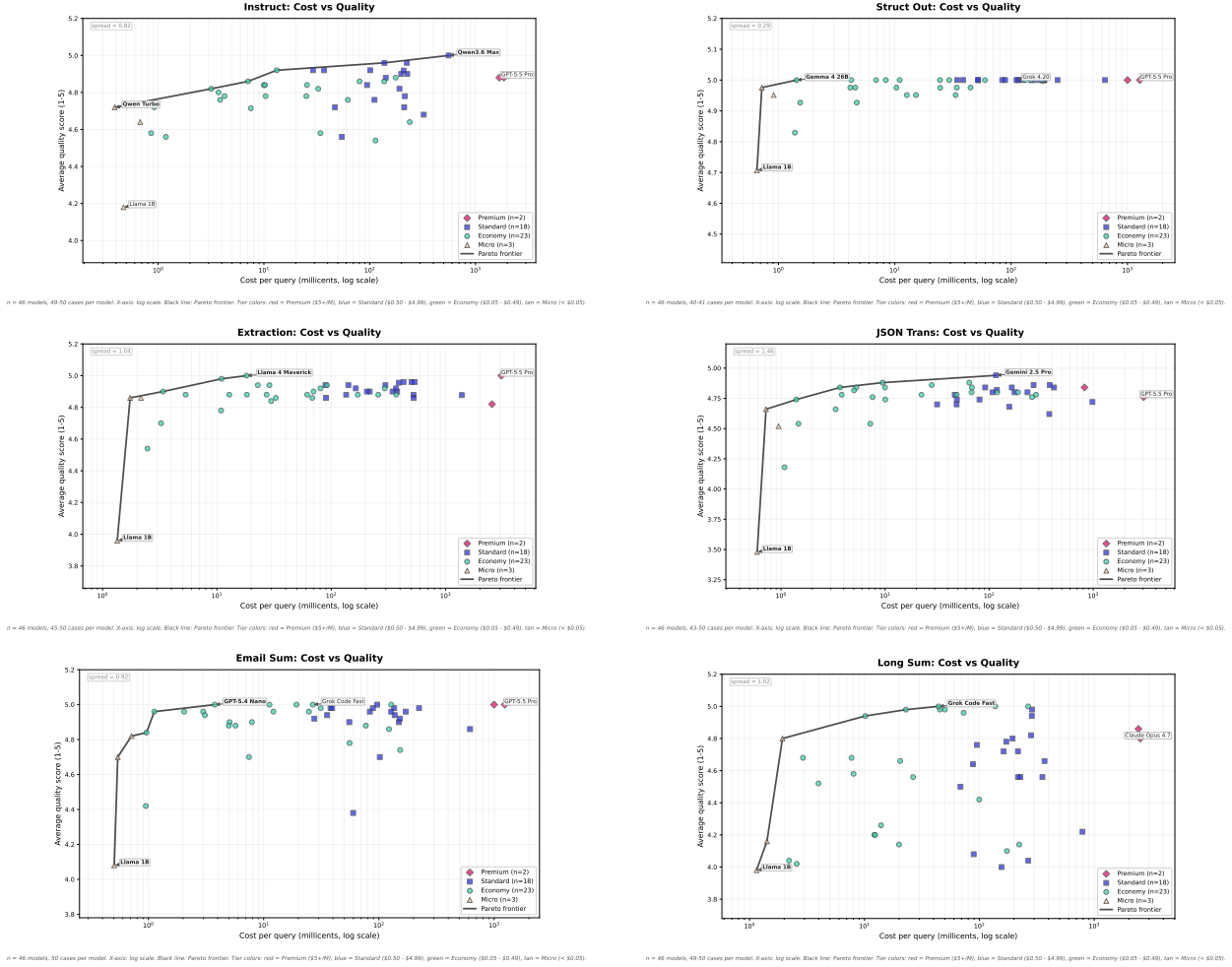
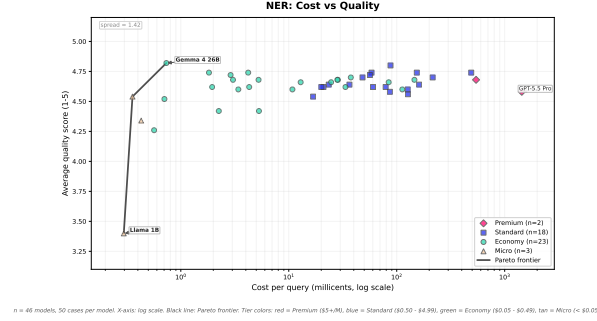


Figure 10: Pareto frontiers (3/4): Structured output and summarization tasks.

H Task Discriminateness

I Verbosity Analysis

J Origin and License Comparisons



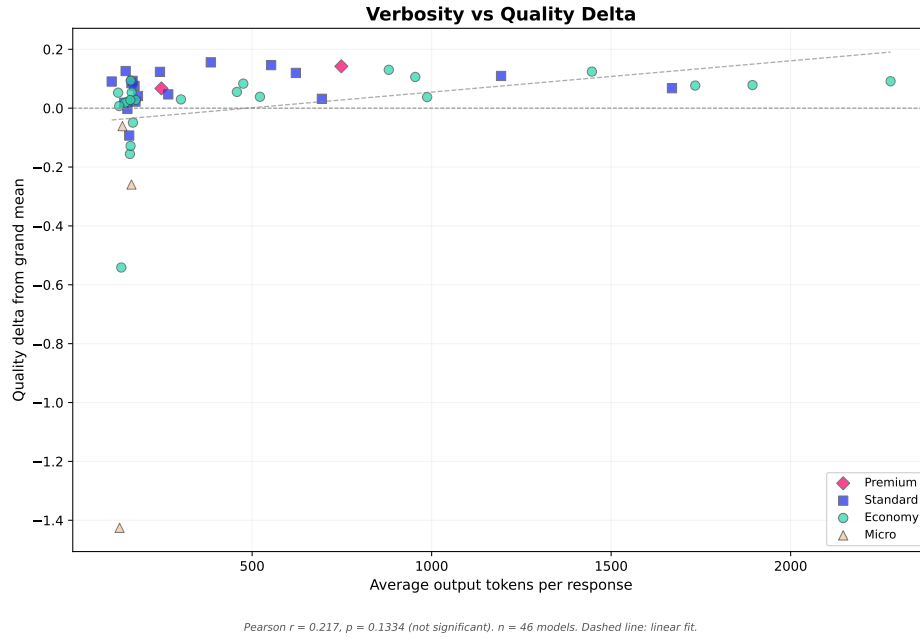


Figure 14: Average output tokens per model versus quality delta from the grand mean. Pearson $r = 0.217$, $p = 0.133$ (not significant). Verbosity does not predict the V1-to-V2 score change, indicating that rubric ambiguity — not output length — drives the format bias.

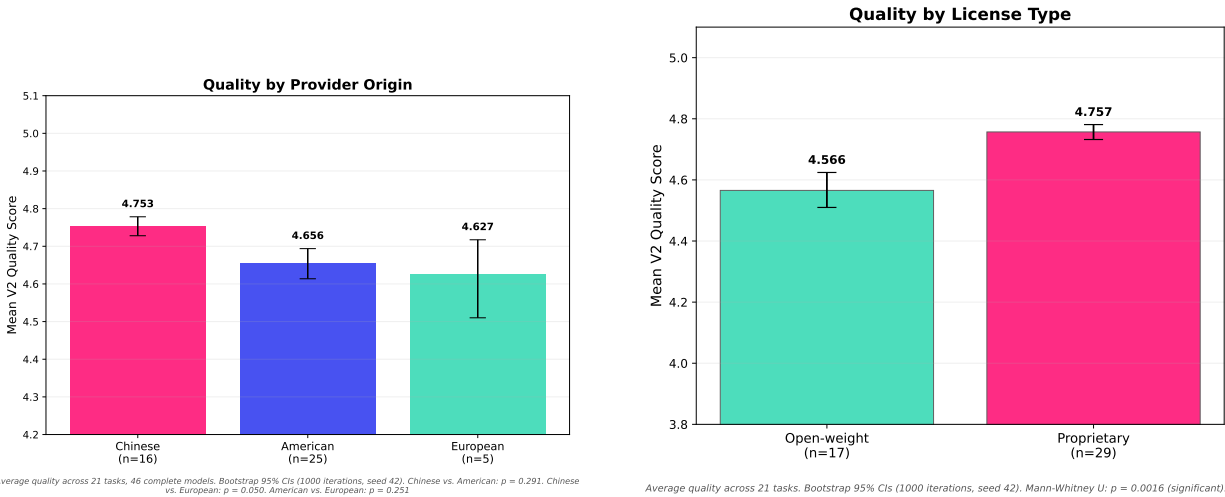


Figure 15: **Left:** Quality by provider origin (bootstrap 95% CIs). No significant difference (Kruskal-Wallis $p = 0.15$). **Right:** Quality by license type (bootstrap 95% CIs). Open-weight scores lower ($p = 0.0016$), driven by sub-3B models.